KNOWLEDGE CHECK

Databricks Fundamental

Course quiz

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Edition: October 2026

The quiz has 40 multiple-choice questions, grouped by module. Choose the single best answer to each question. Check your answers in the quiz of the Workshop Guide app (instant feedback with explanations), or ask the trainer.

MODULE 01

Platform and workspace

1. What are the three main layers of the Databricks lakehouse architecture?

A. Storage (Delta Lake), compute (clusters and SQL warehouses) and governance (Unity Catalog)

B. Bronze, silver and gold

C. Driver, executors and shuffle

D. HDFS, MapReduce and YARN

2. What is a SQL warehouse used for?

A. Compute optimised for SQL queries and BI tools

B. A long-running all-purpose cluster for notebooks

C. A storage container for Delta tables

D. A Unity Catalog object for managing permissions

3. Which magic command runs SQL in a Python notebook?

A. `%python`

B. `%sql`

C. `%run`

D. `%sh`

4. What is the key difference between DBFS and Unity Catalog volumes?

A. DBFS is faster for large files

B. Volumes are governed by Unity Catalog, with access control and lineage; DBFS is a deprecated, ungoverned file system that new workspaces no longer provide

C. DBFS supports the Delta format, while volumes support only Parquet

D. There is no difference

5. Your catalog is `retailhub_anna`. Which path reads `orders/orders_batch.json` from the volume `datasets` in the schema `default`?

A. `dbfs:/mnt/datasets/orders/orders_batch.json`

B. `retailhub_anna.default.datasets.orders_batch`

C. `/Volumes/retailhub_anna/default/datasets/orders/orders_batch.json`

D. `/Workspace/retailhub_anna/datasets/orders/orders_batch.json`

6. What do you pay for when a notebook runs on a classic cluster?

A. Only DBUs; the virtual machines are included

B. DBUs to Databricks plus the Azure virtual machines that run the cluster

C. Only the Azure virtual machines; DBUs are charged only on serverless

D. A fixed monthly fee per workspace

7. A pipeline runs every night as a scheduled job. Which compute choice usually costs the least?

A. An all-purpose cluster that stays on all day, so the job starts faster

B. An all-purpose cluster with auto-termination switched off

C. Job compute or serverless, which starts for the run and stops when it ends

D. The largest SQL warehouse in the workspace

8. What is Genie Code?

A. A chat for business users that answers questions over dashboards and Genie Agents

B. A code formatter that runs when a notebook is saved

C. A library you install with `%pip` to generate PySpark code

D. The AI assistant in notebooks and the SQL editor, formerly Databricks Assistant, which writes, explains and fixes code using your Unity Catalog metadata

9. You ran `display(df)` in a notebook. How do you build a bar chart without writing code?

A. In the result, select +, choose **Visualization**, then the chart type, axes and aggregation

B. Install `matplotlib` with `%pip` and plot the DataFrame

C. Export the result to CSV and open it in Excel

D. Charts are available only in AI/BI dashboards, not in notebooks

MODULE 02

Spark architecture

10. What is the driver responsible for in a Spark cluster?

A. Storing data on disk as the primary storage node

B. Running individual tasks in parallel

C. Coordinating execution: running your code, building the plan and scheduling tasks

D. Compressing data before it is sent to the executors

11. What distinguishes a transformation from an action?

A. Transformations write data; actions read data

B. Transformations are lazy and return a new DataFrame; actions trigger execution

C. Transformations exist only in SQL; actions only in Python

D. Transformations run immediately; actions are lazy

12. When does a shuffle occur?

A. Whenever a new DataFrame is created

B. Only when writing to Delta Lake

C. When data must be redistributed across partitions, typically in wide transformations such as `groupBy` or `join`

D. When `inferSchema` is used on CSV files

13. What does lazy evaluation mean?

A. Spark waits for free disk space before reading files

B. Spark does not execute transformations until an action is called

C. Spark caches every DataFrame after its first read

D. Spark uses a slower but more accurate engine

MODULE 03

ELT and ingestion

14. What is the main drawback of `inferSchema=True` on a large CSV file?

A. The inferred schema is always wrong

B. Spark reads the file twice, once to infer the types and once to load the data

C. Only string columns are supported

D. It does not work with Unity Catalog

15. Which method returns count, mean, standard deviation, minimum and maximum for the columns of a DataFrame?

A. `df.show()`

B. `df.printSchema()`

C. `df.count()`

D. `df.describe()` (`df.summary()` adds percentiles)

16. How does `display(df)` differ from `df.show()`?

A. `display()` writes the DataFrame to a table; `show()` only reads it

B. `show()` works only in SQL cells

C. `display()` renders an interactive table in the notebook with charts and Data Profile; `show()` prints plain text

D. There is no difference

17. ANSI mode is on, which is the default. What do `CAST('abc' AS INT)` and `try_cast('abc' AS INT)` return?

A. `CAST` raises a `CAST_INVALID_INPUT` error; `try_cast` returns `NULL`

B. Both return `NULL`

C. Both return `0`

D. Both raise an error

18. Which call saves the DataFrame `df` as a managed Delta table `bronze.customers` in your catalog?

A. `df.createOrReplaceTempView("customers")`
B. `df.write.mode("overwrite").saveAsTable(f"{CATALOG}.bronze.customers")`
C. `display(df)`
D. `df.cache()`

19. How do you create a temporary view from a DataFrame?

A. `df.saveAsTable("view_name")`
B. `df.createOrReplaceTempView("view_name")`
C. `spark.sql("CREATE VIEW view_name")`
D. `df.registerTable("view_name")`

20. Which approach adds or updates a column using conditional logic?

A. `df.filter()`
B. `df.withColumn()` with `when()` and `otherwise()`
C. `df.groupBy()`
D. `df.dropDuplicates()`

MODULE 04

Delta Lake

21. What does ACID stand for?

A. Accurate, consistent, integrated, durable
B. Atomicity, consistency, isolation, durability
C. Automated, concurrent, incremental, distributed
D. Analytical, columnar, indexed, distributed

22. What is the Delta transaction log (`_delta_log`)?

A. A Spark UI page with query history
B. A directory of CSV files with row-level changes
C. A JSON log that records every change to a Delta table and makes ACID transactions and time travel possible
D. An index that speeds up reads on large tables

23. What does `MERGE INTO` do?

A. Creates a full copy of a table
B. Performs an upsert, updating matching rows and inserting new ones in a single atomic operation
C. Merges two schemas without writing data
D. Combines two transaction logs into one

24. How do you query a Delta table as it was seven days ago?

A. `SELECT * FROM t WHERE _date = date_sub(now(), 7)`

B. `SELECT * FROM t VERSION AS OF 7`

C. `SELECT * FROM t TIMESTAMP AS OF date_sub(current_timestamp(), 7)`

D. `SELECT * FROM t AT (DAYS = 7)`

25. What does `VACUUM` do?

A. Compacts small Parquet files

B. Removes duplicate rows

C. Permanently deletes data files that the current table version no longer references and that are older than the retention period

D. Rebuilds the transaction log

26. An hour ago someone ran a `DELETE` on the wrong rows. Which command brings the table back to its state before that `DELETE`?

A. `ROLLBACK TABLE t`

B. `UNDROP TABLE t`

C. `VACUUM t RETAIN 0 HOURS`

D. `RESTORE TABLE t TO VERSION AS OF n`, where `n` is the version before the `DELETE`

27. You append a DataFrame that has one extra column to an existing Delta table, without any write options. What happens?

A. The extra column is silently dropped

B. The column is added to the table automatically

C. The write fails because the schema does not match (schema enforcement); the `mergeSchema` option adds the column on purpose

D. The table is converted to Parquet

28. What does Databricks recommend for the data layout of new tables, in place of partitioning and `ZORDER`?

A. Liquid clustering (`CLUSTER BY`, or `CLUSTER BY AUTO`); on Unity Catalog managed tables predictive optimization also runs maintenance such as `OPTIMIZE` and `VACUUM` for you

B. Partitioning by every column used in a filter

C. Storing the table as CSV

D. Caching the table on every cluster

MODULE 05

Orchestration with Lakeflow Jobs

29. What does a repair run do?

A. Resets every task and reruns the whole job

B. Reruns only the failed tasks and the tasks downstream of them

C. Sends an alert and cancels the run

D. Rolls back the Delta table changes made by the failed run

30. Which Quartz cron expression runs a job every day at 02:30?

A. `30 2 * * *`

B. `2 30 * * *`

C. `0 30 2 * * ?`

D. `30 2 0 * * ?`

31. A job parameter and a task parameter of a notebook task use the same key. Which value does the notebook read?

A. The task parameter

B. Whichever parameter was created last

C. Neither; the run fails with a conflict

D. The job parameter, which takes precedence

32. Which feature passes parameters to a notebook when a job runs?

A. The `%run` magic command

B. Widgets (`dbutils.widgets`)

C. Unity Catalog parameters

D. Cluster environment variables only

33. The task `validate` publishes a value with `dbutils.jobs.taskValues.set(key="row_count", value=n)`. How does the next task read it?

A. `dbutils.widgets.get("row_count")`

B. `spark.conf.get("row_count")`

C. `dbutils.jobs.taskValues.get(taskKey="validate", key="row_count")`

D. `%run ./validate`

MODULE 06

Unity Catalog

34. What is the three-level namespace in Unity Catalog?

A. `workspace.schema.table`

B. `catalog.database.file`

C. `catalog.schema.table`, where the last level can also be a view, volume or function

D. `metastore.catalog.schema`

35. How must privileges be granted to give a user access to a table?

A. On the table only; higher levels are not needed

B. At the metastore level only

C. Top-down: `USE CATALOG`, then `USE SCHEMA`, then the privilege on the table

D. On the cluster, through IAM roles

36. Which command shows who has which privileges on the schema `gold` of your catalog?

A. `SHOW GRANTS ON SCHEMA my_catalog.gold`

B. `DESCRIBE SCHEMA my_catalog.gold`

C. `SHOW TABLES IN my_catalog.gold`

D. `LIST PRIVILEGES my_catalog.gold`

37. Which groups can be granted privileges on Unity Catalog objects?

A. Workspace-local groups created with SQL `CREATE GROUP`

B. Account groups, created in the account console or in the workspace settings under Identity and access

C. The workspace system group `users`

D. Any group that exists in the workspace

38. A participant has `ALL PRIVILEGES` on a catalog owned by the trainer, but their `GRANT SELECT ... TO` a group fails. Why?

A. `GRANT` can be run only from a SQL warehouse

B. `ALL PRIVILEGES` must be granted at the metastore level before grants are allowed

C. `ALL PRIVILEGES` does not include `MANAGE`; only the owner, a metastore admin or a holder of `MANAGE` can grant privileges

D. Groups can receive privileges only on tables

MODULE 07

Genie Agents, optional demo

39. What is a Genie Agent (formerly Genie space)?

A. A GPU cluster for machine learning models

B. A natural-language interface, configured on chosen Unity Catalog tables, that answers questions about the data with generated SQL

C. A shared area for editing notebooks together

D. A dashboard for monitoring pipeline health

40. When is a Genie Agent not the right tool?

A. When a business user wants to explore data without writing SQL

B. When you need a strictly formatted report with approved KPIs

C. When an analyst wants to ask ad-hoc questions about sales

D. When you want to check a trend before a meeting